

A formula-level reading of the Spiral-Time neurodynamic model

Krüger–Feeney–Wende (*Medinformatics* 2026), the foundational time-axis, and the FFGFT conversion-as-measurement reading

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Abstract

This document is a reproducible, formula-level reading of *Information-Theoretic Modeling of Neural Coherence via Triadic Spiral-Time Dynamics: A Framework for Neurodynamic Collapse* (Krüger, Feeney & Wende, *Medinformatics* 2026, DOI 10.47852/bonviewMEDIN62029043). Every verdict is produced by a short runnable script. The reading separates what holds (an interpretable phase-deviation feature; genuine octonion non-associativity) from what is mislabeled (the triadic algebra is quaternionic, not bicomplex), internally inconsistent (two incompatible definitions of $\Delta\Phi$), not derived (the 0.40 threshold; the claimed spirals and limit cycles), or not reproducible from the paper (the reported $r \approx 0.96$). Underneath these sits *one* foundational issue: the operator $\psi(t) = t + i\varphi(t) + j\chi(t)$ is built on physical, linear time as its base axis – the decompactified, post-measurement coordinate – so its memory is generic by construction. The FFGFT reading makes this precise: the system is compact and carries memory. By Doc. 270 the decompactification to SI time is a mode-preserving embedding (lossless), and FFGFT does carry framework-specific SI observables (masses/Koide, CHSH $\sim \xi$, D_f , L_0 ; Doc. 270/249); what removes the recurrence is not the SI conversion but the finite, non-periodic observation window plus decoherence. The sharp, narrow consequence: the coherence-sector recurrence is not *resolvable* on a warm, finite-window SI-time signal such as EEG. This is a reviewer note addressed to the authors; the intent is the falsification-forward standard the HLV/HICE side already applies.

.1 Occasion and method

The Medinformatics paper applies the HLV Spiral-Time operator to neural coherence and proposes the Information–Coherence–Energy (ICE) stability principle and a $\Delta\Phi$ -based “neurodynamic collapse” boundary. Rather than argue from impression, each mathematical claim here is checked by computation. The scripts are `test_algebra.py`, `test_deltaphi.py`, `test_dynamics.py`, `test_compactification.py` and `make_figures.py`; fixed seeds throughout. Findings split into: *holds*, *mislabeled*, *internally inconsistent*, *not derived*, *not reproducible from the paper*, and *speculative premise*. One issue is foundational – the time-axis the operator is built on – and is taken first, because findings on $\Delta\Phi$ and on the dynamics are its symptoms.

#	Claim	Section	Status
0	Base time-axis $\psi(t) = t + i\varphi(t) + j\chi(t)$ on physical linear time	2.1, 2.2.1	Foundational – operator on the post-measurement axis (subsumes 3 & 5)
1	Triadic time “reduced bicomplex/octonionic plane”	2.1	Mislabeled – it is the quaternions
2	Octonionic operator; non-assoc. “explains psychology”	2.10	Math holds, claim overreaches
3	$\Delta\Phi$ instability functional	2.6 vs 12/15	Internally inconsistent (two definitions)
4	Critical $\Delta\Phi_c \approx 0.40$, “not a free parameter”	2.5, 2.6.2	Not derived (free parameter)
5	Healthy spirals / trauma limit cycles	2.2.3, 2.9	Not derived from the equations
6	$r \approx 0.96$, $\Delta\Phi_c \approx 0.40$ “across subjects”	3.4	Not reproducible from the paper
7	Axonal biphoton entanglement coupling Γ_{axon}	2.7–2.8	Speculative premise as established

.2 The base time-axis (foundational)

This is the most consequential choice in the paper, and the one under which findings 3 and 5 are downstream symptoms. The Spiral-Time operator

$$\psi(t) = t + i\varphi(t) + j\chi(t) \quad (1)$$

takes physical, linear time t as its scalar base and writes φ, χ as functions of that same t . The paper is explicit that t is physical: U_1 is defined as “linear physical time,” tracking external chronology (§2.2.1). That fixes the representation in which every later memory and holonomy claim is made.

Unit-independent. For $t + i\varphi + j\chi$ to be one algebraic object the three components must be the same kind of quantity. The paper asserts dimensionlessness (§2.1.2) yet simultaneously calls t physical time and chronology (§2.2.1); φ and χ are internal phase / identity coordinates. Adding a physical time to internal phases in one quaternionic sum is dimensionally incoherent unless all three are first reduced to a common dimensionless measure. (Since i, j anticommute the object is quaternionic – t is the real part of a quaternion whose cross-term $ij = k$ is never tracked; see §.3.)

Deeper. A framework-specific memory lives on the compact internal time-circle S^1 : a discrete spectrum, an exact recurrence (in FFGFT the coherence-sector back-flow 5.125, Doc. 282). Writing $\varphi(t), \chi(t)$ as single-valued functions of physical linear t flattens S^1 onto $\mathbb{R}$ – it performs the projection $S^1 \rightarrow \mathbb{R}$ before any dynamics, and a single-valued function over linear time has no compact periodicity, so the recurrence is not representable on this axis at all. By the conversion-as-measurement principle (§.11), committing to physical time is itself the disturbing, information-discarding projection – no apparatus needed. The notation has silently taken that step, hiding the one operation – the conversion/measurement from compact internal time to physical SI time – that decides whether a memory claim is framework-specific or generic.

Consequence. The operator is, by construction, post-measurement: whatever memory it carries on physical (EEG) time is generic. This is reproducible (`test_compactification.py`): the discrete-spectrum return signal has back-flow 11.5 on the compact horizon, which a finite physical read-out window drives to 0.000, while the six lines never resolve ($2\pi/T_{\text{win}} \gg$ line spacing 0.091) – the read-out is a non-invertible projection. The same root explains the two downstream findings: a Berry-phase holonomy (Eq. 12/15, §.5) needs a closed loop – a compact cycle – and a limit cycle (§.7) needs non-gradient closed orbits; neither is supported on a linear physical-time axis.

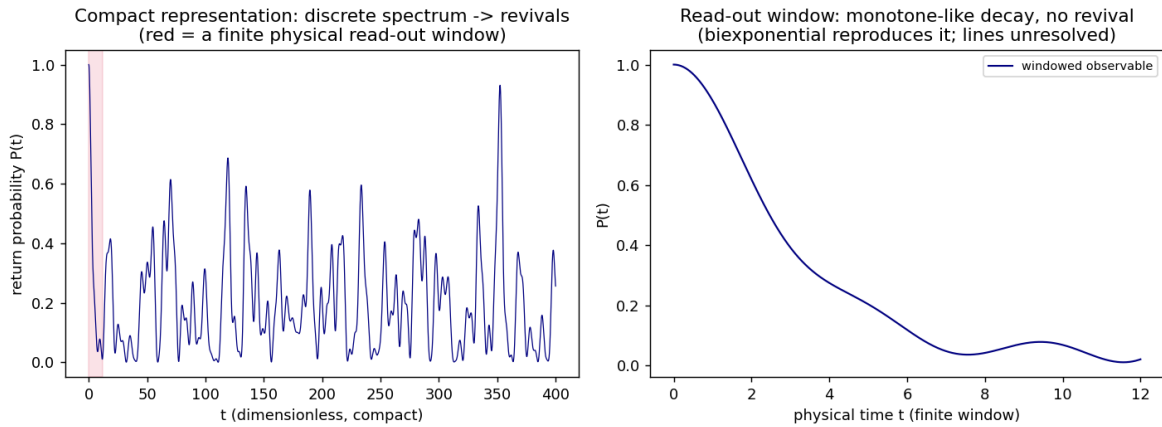


Figure 1: Read-out as a non-invertible projection. Left: on the compact representation the discrete spectrum produces revivals (back-flow $\gg 0$); the red band marks a finite physical read-out window. Right: inside the window the observable decays without revival and the discrete lines are unresolved.

Constructive fix. Define the operator on the compact, dimensionless internal time (cyclic), where the discrete spectrum and recurrence live, and let physical SI time enter only through an explicit, declared conversion – which is precisely what projects out the framework-specific part. Either the operator stays internal/compact (no direct EEG claim) or it is physical/SI (generic memory). $\psi(t) = t + i\varphi(t) + j\chi(t)$ tries to be both. The minimal change is to declare two distinct times and write the conversion between them explicitly as the measurement it is.

.3 Triadic time algebra – mislabeled

The stated relations $i^2 = j^2 = -1$, $ij = -ji$ are exactly the imaginary quaternions: setting $k := ij$ gives $k^2 = -1$, and $\{1, i, j, k\}$ closes into $\mathbb{H}$, which is associative and non-commutative (`test_algebra.py`, A). A *bicomplex* plane requires the two imaginary units to commute ($ij = ji$); here $ij = -ji$, so the algebra is *not* bicomplex. A three-term operator $t + i\varphi + j\chi$ lives inside a quaternion sub-algebra; octonions are not required for it. The correct name is quaternionic.

.4 Octonionic structure – math holds, interpretation overreaches

Octonion non-associativity is real: the associator $(e_1 e_2) e_4 - e_1 (e_2 e_4) = 2e_7 \neq 0$ (Cayley–Dickson), while associators on a quaternionic triple vanish (`test_algebra.py`, B). That is correct

mathematics. The further step – that this “directly matches empirical psychology” and is “the first physics-based explanation for psychological nonlinearity” – is an interpretive analogy, not a derivation: no psychological quantity is computed from or tested against the algebra, and the eight basis directions are assigned meanings by stipulation.

.5 $\Delta\Phi$ – two incompatible definitions under one symbol

- **Definition 1 (§2.6):** $\Delta\Phi = \alpha|\Delta S| + \beta|\Delta I| + \gamma|\Delta C|$, convex weights, each component in $[0, 1]$ – a unitless weighted- L^1 index in $[0, 1]$. It is linear in each component, so it has no curvature and no intrinsic bifurcation; the level set $\Delta\Phi = c$ is a flat polytope facet for every c .
- **Definition 2 (Eq. 12, 15):** $\Delta\Phi = \oint (\nabla \times A) \cdot da \approx \text{Im} \oint \langle \psi | \nabla \theta | \psi \rangle d\theta$ – a Berry-phase holonomy, an angle in radians ranging over $0 \dots 2\pi$, with no reason to lie in $[0, 1]$ or near 0.40 (a loop with $|\gamma| = 0.40$ rad is a tiny $\sim 29^\circ$ cone).

A normalized deviation index and a radian holonomy are different mathematical objects. Calling both “ $\Delta\Phi$ ” and assigning both the same critical value 0.40 is an equivocation (test_deltaphi.py, C/D).

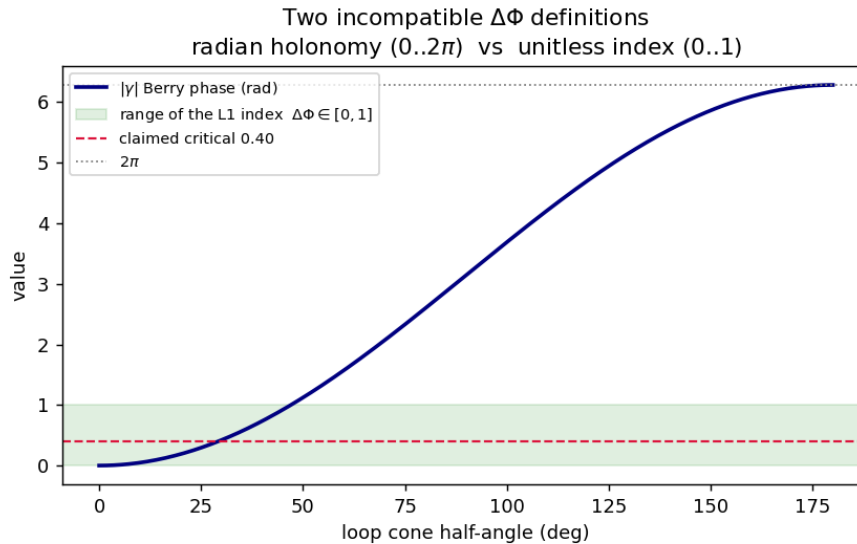


Figure 2: The two $\Delta\Phi$ definitions are incompatible objects: a radian holonomy ($0 \dots 2\pi$) versus a unitless index ($0 \dots 1$).

.6 The 0.40 threshold – not derived

§2.5 states the value is “not imposed as a free tuning parameter” and that the operator structure “enforces” a bounded instability surface. But the L^1 functional has no intrinsic critical point (it is linear); the only structural statement available is the trivial bound $0 \leq \Delta\Phi \leq 1$ and the symmetric reference $\Delta\Phi = d$ (so $1/3$ at $d = 1/3$). §2.6.2 itself attributes the 0.40-vs-0.333 gap to “dataset-specific weighting asymmetry ... and normalization,” and at equal deviation $d = 0.40$ every convex weight set returns 0.40 (test_deltaphi.py, C/E). So 0.40 is an empirical/contour value, not a structurally enforced bifurcation.

.7 Spirals and trauma limit cycles – not derived

The only dynamical mechanism given with explicit potentials (§2.4) is the gradient-flow part of the master equation, $\partial_t \Psi = -\nabla V_{\text{DLHR}}(\Psi)$. For any gradient flow,

$$\frac{dV}{dt} = \nabla V \cdot \frac{d\Psi}{dt} = -|\nabla V|^2 \leq 0, \quad (2)$$

so V is a strict Lyapunov function and the system cannot have periodic orbits, limit cycles, or sustained spirals – it relaxes to fixed points (theorem). Numerically (`test_dynamics.py`, F): across 40 trajectories $V(t)$ is monotonically decreasing (0 violations) and 0 closed orbits are found. The phenomenology “healthy cognition → stable inward spirals,” “trauma loops → limit cycles” therefore does not follow from the specified dynamics; it would require the advection terms $\lambda_\varphi \partial_\varphi$, $\lambda_\chi \partial_\chi$ or W_{oct} , which are never given concretely. (The cubic M^3 term also makes V unbounded below.)

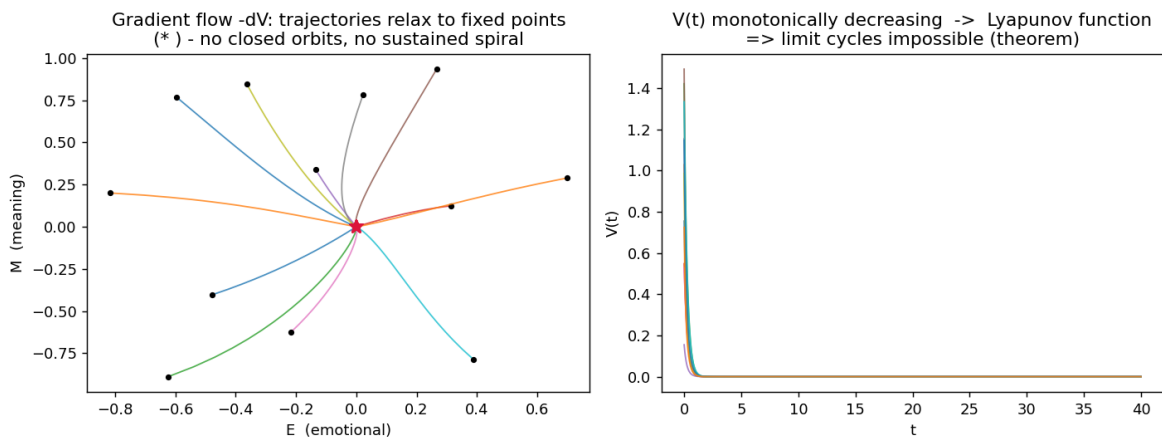


Figure 3: Gradient flow with the paper’s DLHR potentials: trajectories relax to fixed points (no closed orbits, no sustained spiral) and $V(t)$ decreases monotonically, so limit cycles are impossible.

.8 Empirical claims – not reproducible from the paper

The reported $r \approx 0.96$ and $\Delta\Phi_c \approx 0.40$ “across subjects” (§3.4) cannot be recomputed from the paper, which gives no data table. Reproduction needs the external dataset (Zenodo 10.5281/zenodo.17990696) and the validation notebook. A fair test would require the dataset plus a pre-registered protocol with the nulls the authors already use on the HICE side (time-shuffle, present-state-matched, phase-randomized).

.9 Axonal biphoton entanglement – speculative premise stated as established

$\Gamma_{\text{axon}} = g e^{i\varphi} E_{12} + g' e^{i\chi} E_{23}$ (§2.8) is built on “experimental findings” of biphoton entanglement and long-range coherence in myelinated axons, presented as settled research. Entanglement in warm, wet, myelinated axons is not established physics. This is not a formula error, but it is a load-bearing empirical premise asserted as fact and should be flagged as hypothesis.

.10 A defensible core

Stripped of the superstructure there is a sound, modest object: $\Delta\Phi = |\omega(t) - \omega_{\text{memory}}(t)|$ – the deviation of instantaneous phase velocity from a short-term phase-memory baseline – is an interpretable, lightweight time-series instability feature, and as a signal-analysis operator (the framing of the companion respiratory paper) it is legitimate. The problems are confined to the HLV/octonion/consciousness superstructure, the $\Delta\Phi$ redefinition, and the un-derived threshold and dynamics.

.11 An FFGFT reading: conversion as measurement

Temporal recursion does carry memory, and it maps onto FFGFT's own structure (the recursion operator $\mathcal{R}$; the memory kernel as the Fourier transform of the connection-Laplacian spectral density, Doc. 282). The following is offered as a constructive bridge and an explicit conjecture, not a result.

FFGFT's memory has two faces. On the overdamped / relaxation observable the kernel is a positive multi-exponential, and a generic biexponential reproduces it (residual $\sim 10^{-3}$): non-identifiable, Stufe-0. On the coherence observable $c(t) = e^{-\Gamma(t)}$ – its home sector – the discrete spectrum produces a genuine revival: back-flow 5.125, which generic decaying kernels cannot reproduce (back-flow 0). So the memory is discriminating, but only on the coherence sector and only when *both* a discrete spectrum *and* preserved coherence (small Γ) hold.

FFGFT is fundamentally compact (T^4 , discrete spectrum), and in that dimensionless representation the system always has memory. It is important to locate the loss precisely, because the conversion to SI is *not* where it happens. By Doc. 270 the Type I duality decompactification $S_m^1 \rightarrow \mathbb{R}_t$ (governed by $\tilde{T} \cdot m = 1$) is a *mode-preserving embedding* – lossless; the periodicity merely stops being directly visible and survives as the discrete spectrum and as physical observables (the revival time T_{rev} , the CHSH deviation $\sim \xi$). A bare units rescaling (by $\hbar, c$) is likewise invertible and loses nothing. The genuinely lossy operations are three and distinct: the *reverse* compactification $\mathbb{R} \rightarrow S^1$ (non-periodic parts do not descend), the spatial projections $D3 \rightarrow D0$ (Doc. 270), and – for any read-out – the restriction to a *finite, non-periodic observation window plus decoherence*. It is this last step, not the unit conversion, that keeps the local short-time behaviour and discards the global recurrence, exactly as a measurement selects a basis and drops the coherences (Test G). A physical measurement is the same window-and-decoherence projection at its strong-coupling end. This is consistent with what the corpus already records: the vacuum energy is present compactified, visible decompactified, with $c, \hbar c$ as SI conversion factors (Doc. 263) – the decompactified view is not less true, only differently resolved.

The reading therefore both supports the modest core (recursion does carry memory) and bounds the claim precisely. The framework-specific signature does survive into SI in the right channels – mass reading, CHSH $\sim \xi, D_f, L_0$ (Doc. 270/249). What does *not* survive is the coherence-sector recurrence read on a warm, finite-window SI-time observable: there the window and decoherence have already discarded the global recurrence, so the very observable the clinical sections use – SI-time EEG – is post-measurement in exactly that sense, and is the regime where the FFGFT harness (Doc. 282/283) shows the memory to be generic, not a fingerprint. This is also why the base-time-axis choice (§3) is decisive: the operator is written on that finite, post-measurement axis from the start.

.12 Reproducibility

- `test_algebra.py` – quaternion closure + bicomplex refutation (A); octonion associator (B).
- `test_deltaphi.py` – L^1 bound/linearity (C); Berry-phase range (D); 0.40 as free knob (E).
- `test_dynamics.py` – gradient-flow Lyapunov / no-limit-cycle test (F).
- `test_compactification.py` – read-out as a non-invertible projection: compact back-flow vs windowed (G).
- `make_figures.py` – the figures used above.

The figures `fig_deltaphi.png`, `fig_dynamics.png` and `fig_compactification.png` are produced by these scripts; all use fixed seeds.